ELEC60003
ELEC70043

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MSc and EEE PART III/IV: MEng, BEng.and ACGI

ANALOGUE INTEGRATED CIRCUITS AND SYSTEMS

Tuesday, 5 December 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : C. Toumazou
Second Marker(s) : P. Georgiou

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1. Figures 1 (a, b c, and d) show four popular biasing schemes typically used in analogue integrated circuits.

   (a) Explain the function of each of the circuits in Figures 1 (a, b, c and d) and derive expressions for the constant output parameter in each case, clearly indicating component design requirements and any approximations you have made. You may ignore bulk effects in the CMOS circuits.

   [16]

   (b) Draw an improved version of the circuit shown in Figure 1(c) which also guarantees the voltage V_z is received by the output load and explain its operation.

   [4]

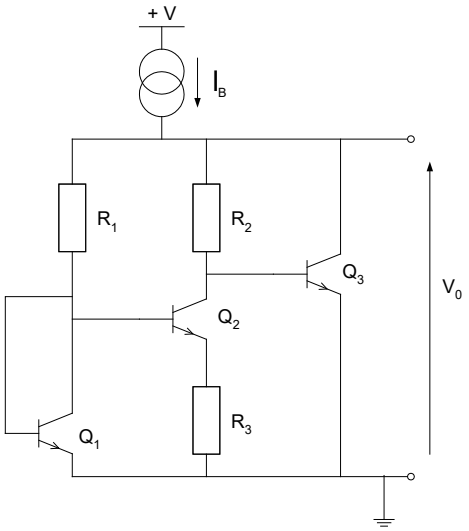


Figure 1(a)

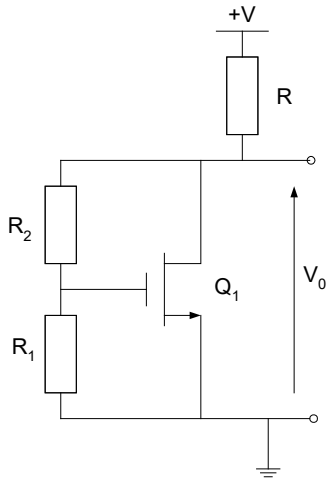


Figure 1(b)

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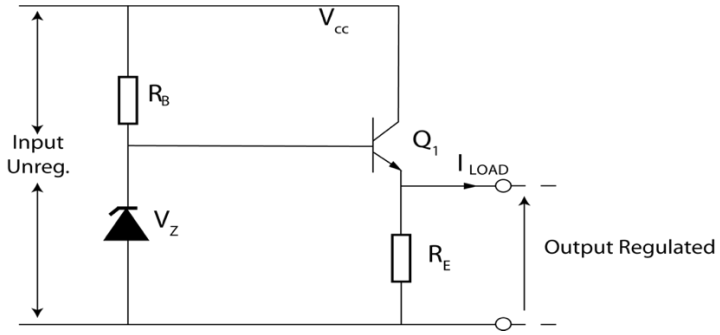


Figure 1(c)

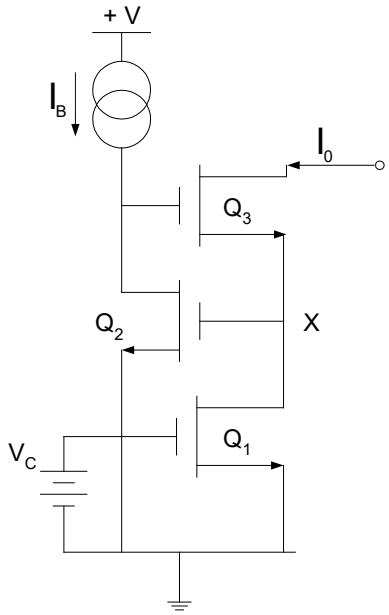


Figure 1(d)

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2. (a) Figure 2(a) shows a commonly used circuit for providing a stable current reference. The circuit has the following values I_O = 5μA, R_1 = 14.4KΩ, I_{IN} = 1mA, R_2 = 26.5KΩ. R_2 is a diffused resistor with a temperature coefficient of +1500ppm/°C at T=300K. The output current sensitivity to the supply voltage V_{cc} is given by:

$$S_{V_{CC}}^{Io} = \frac{1}{1+\frac{I_O R_2}{V_T}} S_{V_{CC}}^{I_{IN}}$$

i) Derive the term S_{VCC}^{Iin} and thus the output current sensitivity.

[4]

ii) Calculate the change in I_o if VCC where to change by 20%.

[4]

iii) The fractional temperature coefficient of the circuit is given by:

$$TC_F = \left(\frac{1}{I_O}\right)\left(\frac{\delta I_O}{\delta T}\right) = \left(\frac{1}{V_T}\right)\frac{\delta V_T}{\delta T} - \left(\frac{1}{R_2}\right)\frac{\delta R_2}{\delta T}$$

Calculate the value of TC_F stating assumptions where necessary.

[4]

(b) Figure 2b shows an improvement to the circuit in Figure 2(a).

i) Describe two improvements this circuit has for generating a constant bias.

[4]

ii) Calculate the fractional temperature coefficient for this circut at room temperature, given that *R* is a polysilicon resistor with a temperature coefficient of 1500 ppm/°C, the temperature coefficient of V_{BE} to be *-2.5mV/°C* both at T=300K. State all assumptions.

[4]

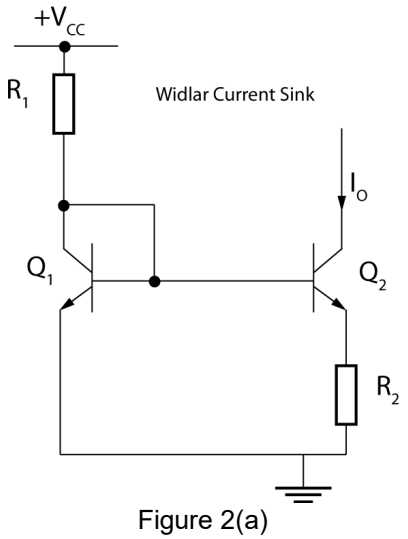


Figure 2(a)

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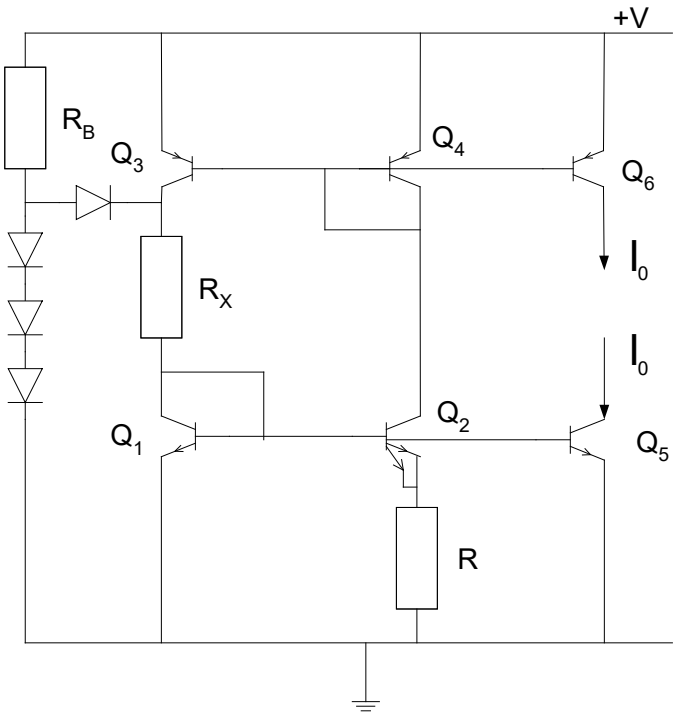


Figure 2(b)

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3. (a) Figure 3 shows a two-stage CMOS op-amp. Estimate the low-frequency differential voltage gain and gain-bandwidth product of the amplifier. Aspect ratios of all devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below. What is the main advantage and disadvantage of a single-stage compared to a two-stage op-amp ?

[10]

(b) Show how the output transistors can be cascoded to increase the voltage gain. State two advantages and two disadvantages of a cascoded output stage.

[10]

CMOS TRANSISTOR PARAMETERS

MODEL PARAMETERS	Kp ($\mu A/V^2$)	λ (V^{-1})	V_{T0} (V)
PMOS	20	0.03	-0.8
NMOS	30	0.02	1.0

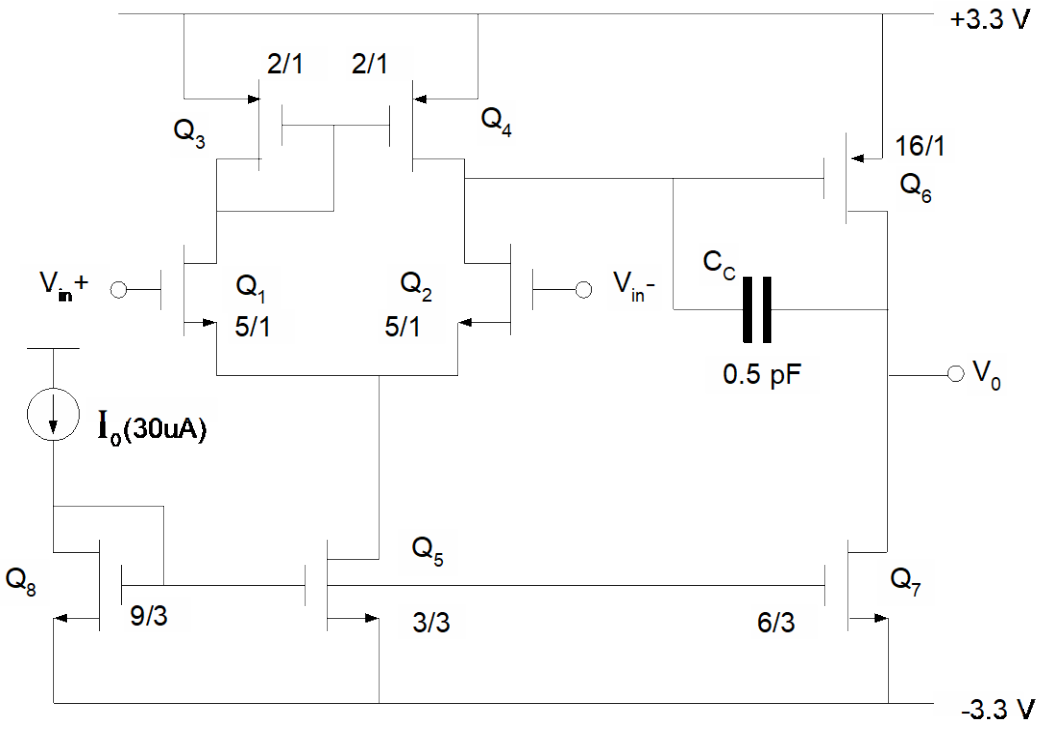


Figure 3.

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4.(a) Under which operating conditions does the MOSFET of Figure 4.1 realise a linear floating resistor between terminals *A* and *B* ? Explain why the bulk is not connected to the source.

[4]

(b) Discuss three sources of non-linearity in the single MOSFET resistor realisation of Figure 4.1 and suggest one suitable circuit design to help to eliminate one or more of these non-linear terms. Show all necessary circuit analysis to confirm your design.

[6]

(c) Show how the two transistor differential integrator shown in Figure 4.2 can be modified to create a double differential integrator. Sketch the new circuit and derive the time constant.

[4]

(d) The integrator in Figure 4.2 is to be used in a biomedical application as part of an ECG patch to measure rhythms of the heart. The output voltage needs to be quantised with low noise and high dynamic range. Sketch a suitable ADC which suits this purpose and explain its operation.

[6]

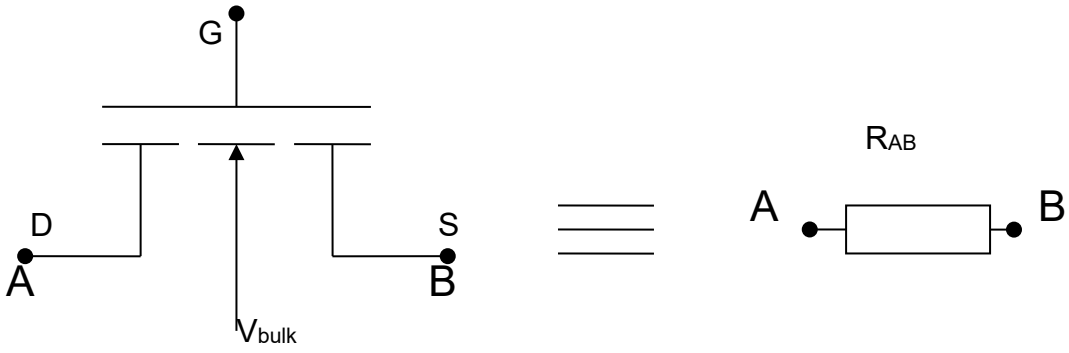


Figure 4.1

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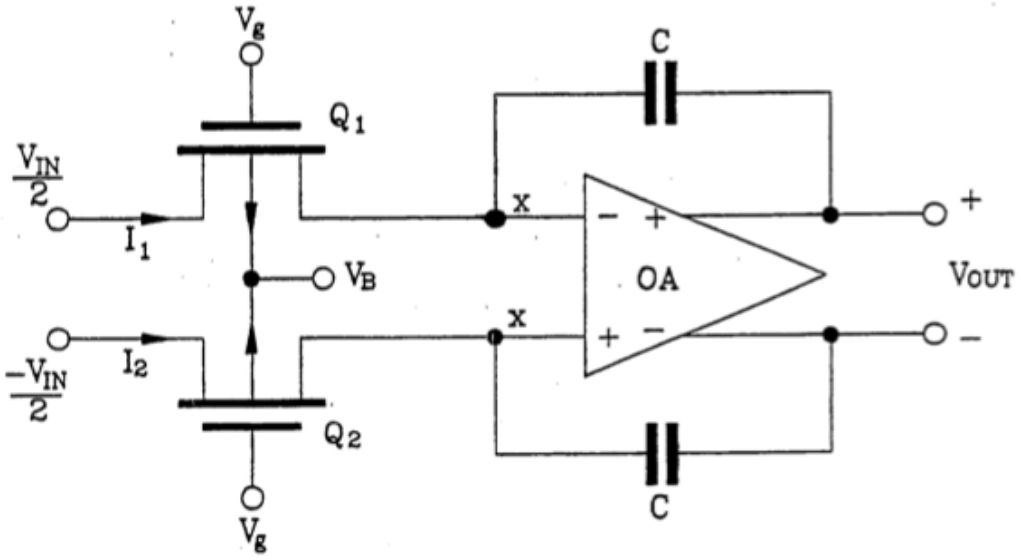


Figure 4.2

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5. Figure 5 shows the basic design of an analogue sampled-data precision integrator.

   (a) Derive an expression for the transfer function of the integrator. Assume that the integrator is driven by non-overlapping clocks and that the switches are ideal.

   [10]

   (b) (i) Sketch the basic design of a 3rd-order Chebyshev low pass switched-capacitor ladder filter.

   The filter is to have a cut-off frequency of 5kHz. Assume a clocking frequency of 100 kHz. The values of integration capacity for the capacitor based sections are 5.08pF and inductive section is 3.49pF. All other switched capacitors are 1pF.

   (ii) From the circuit, estimate normalised passive component values for the original double-terminated LC prototype of the filter. All values should be normalised to 1 rad/s. You may assume that the clocking frequency is so high that the integrators can be assumed lossless.

   [10]

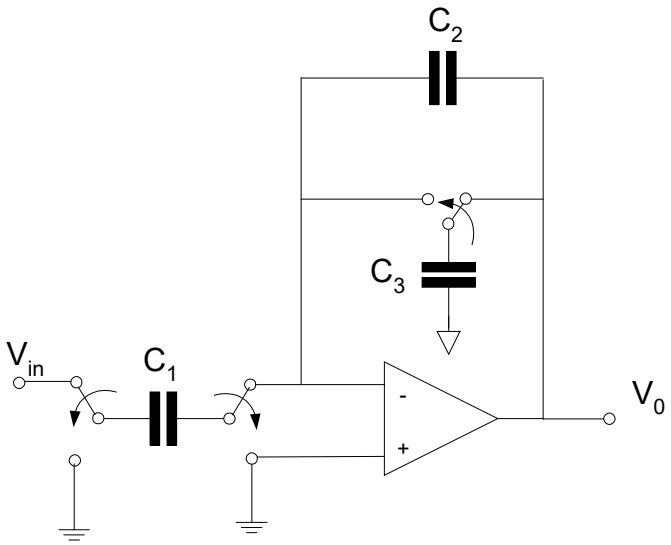


Figure 5.

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6. (a) Sketch the circuit of a single-stage differential to single ended output high frequency operational amplifier that does not require miller compensation.

[5]

(b) When matching transistors special care is taken with the layout. Discuss what is meant by common centroiding when matching two transistors.

[5]

(c) For fully differential output op-amps explain what is meant by common mode feedback using a diagram where necessary.

[5]

(d) Sketch the circuit of a differential operational amplifier that has a single dominant pole and uses common mode feedback to stabilise the output DC voltage.

[5]

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